

The Internet Already Knows I'm Pregnant



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Intro

Kashmir

- Journalist
- Privacy Expert
- New Mother

Cooper

- Hacker
- Privacy Activist
- Cares about privacy issues for people with wombs

Skill sets

Kash



Cooper



Anti-Choice Groups Use Smartphone Surveillance to Target 'Abortion-Minded Women' During Clinic Visits

May 25, 2016, 6:52pm Sharona Coutts

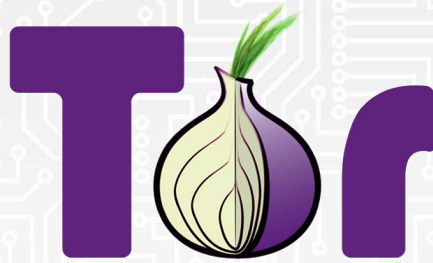
Women who have visited almost any abortion clinic in the United States have seen anti-choice protesters outside, wielding placards and chanting abuse. A Boston advertiser's technology, when deployed by anti-choice groups, allows those groups to send propaganda directly to a woman's phone while she is in a clinic waiting room.



Geo-fencing technology can be deployed by anti-choice groups to send propaganda directly to a woman's phone while she is in a clinic waiting room.

 Kiva Bay for Rewire

How to hide a baby bump



Retailers want to know who's preggers

FEB 16, 2012 @ 11:02 AM 3,192,470

2 FREE Issues of Forbes

How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

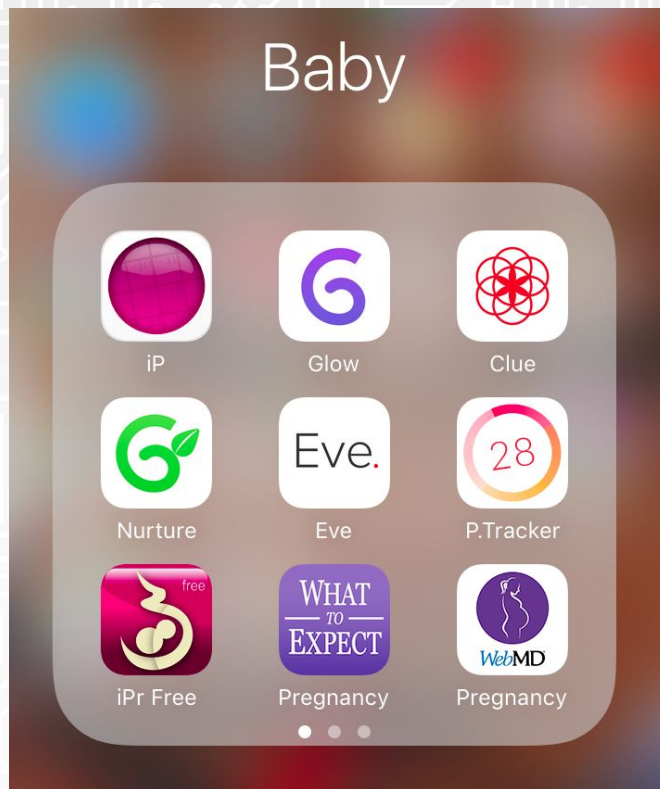
 **Kashmir Hill**, FORBES STAFF 
Welcome to The Not-So-Private Parts where technology & privacy collide [FULL BIO](#) 

Every time you go shopping, you share intimate details about your consumption patterns with retailers. And many of those retailers are studying those details to figure out what you like, what you need, and which coupons are most likely to make you happy. [Target](#) TGT +0.91%, for example, has figured out how to data-mine its way into your womb, to figure out whether you have a baby on the way long before you need to start



Target has got you in its aim

How not to hide it



What apps we tested

pTracker

Glow

nurture

clue

eve

what to expect

pregnancy+

webmd baby

pinkpad

flo

my calendar (Book)

mycalendar (Face)

fertility friend

get baby

babypod

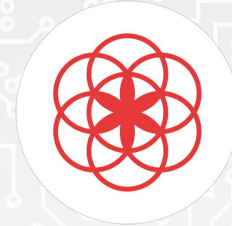
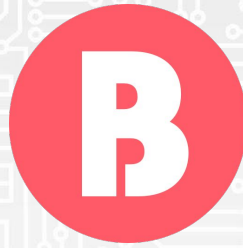
BabyBump

Ovia

The Bump

Maya

The logos





Why people use these apps

← Me Your Journey

Glow supports your personal health tracking from period to parenting. Choose your experience here!

GLOW

Avoiding pregnancy
Gain insights on your body and fertility patterns to avoid pregnancy.

Trying to conceive
Receive accurate predictions on your fertility to boost your chances of pregnancy.

Fertility treatments
Manage the details of your medications, appointments, and results.

22 weeks & 5 days
Your baby is the size of a papaya

S M T W T

121 DAYS TO GO

 Birth Stories

horrific story!
Everything in my pregnancy was so easy! No morning sickness,...



👍 362 💬 26 Last reply 2 days ago

 Baby Bump Selfies

37 weeks exactly 🤗💙👶
Can't wait to meet my love 💙💙💙



👍 51 💬 2 Last reply 12 hours ago

 Period Talk

Why is it greenish black

TMI

What these apps ask about

20 Today Done

S M T W T F S

16 17 18 19 20 21 22

< [Smiley Face] [Waves] [Lightning Bolt] >

FLUID i

EGG-WHITE STICKY

CREAMY ATYPICAL

Jul 20, 2017 Cycle Day 1 Period

See all that your data reveals! See all that your data reveals!

Daily Log

Sex Menstruation Symptoms

Moods Sleep All logs

Sex

Had sex Multiple sessions

No sex

Save

Physical symptoms

Symptoms Mild Moderate Severe

Migraine

Nausea

Pain during sex

Pelvic pain

Sex drive

Sick

Sore breasts

Vaginal pain

Done

Did you take your [prenatal vitamin](#)? ☒ ☐

Did you do [kegels](#)? (set of ten 3x a day) ☒ ☐

[Vaginal discharge](#)? ☒ ☐

Physical

Update your [weight](#) 135 lbs

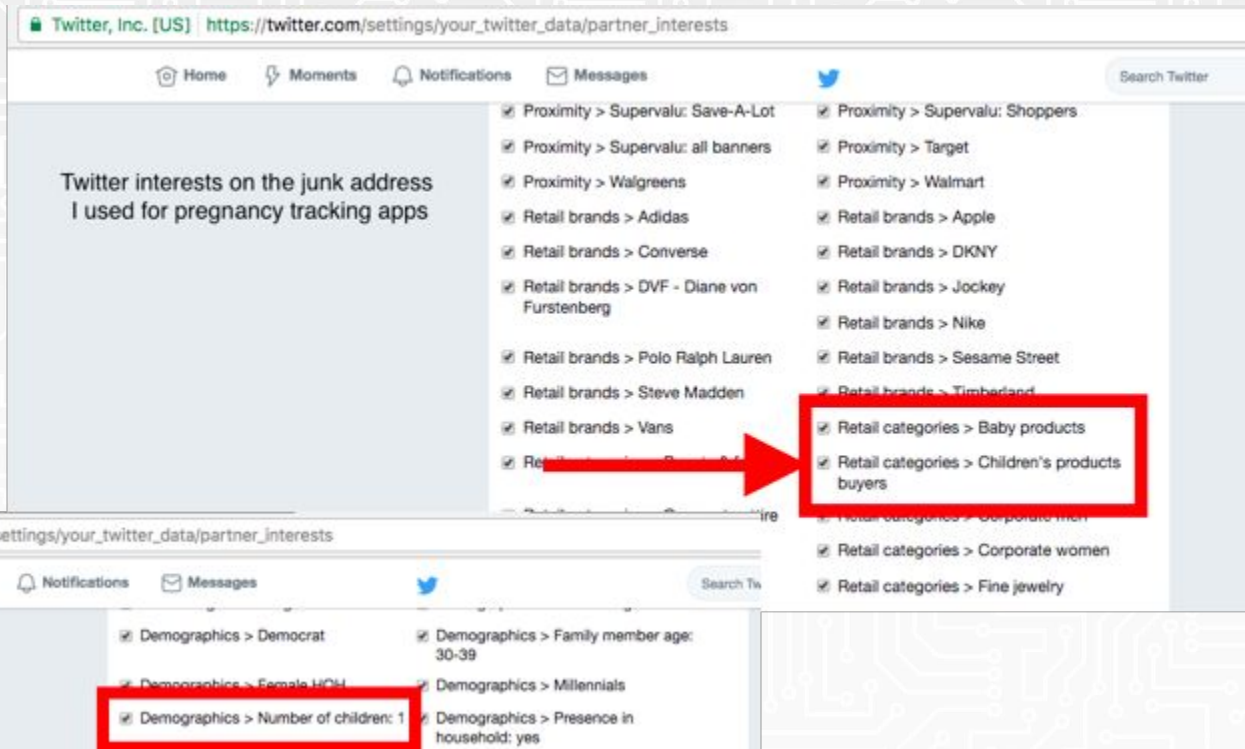
[Did you exercise](#)? ☒ ☐

Did you [drink water](#)? ☒ ☐

Your [sleep duration](#) Choose

Who figured out I was pregnant

Twitter



These people. (My inbox at 6 months)

Focused Other

Pottery Barn Kids	It's all on SALE! 20% OFF everything at our Friends & Family Event (furniture, too!) • Plus:	10/5/2016
Pottery Barn Kids	For our friends and family: 20% OFF everything (even furniture!) – in stores and online •	10/5/2016
WhatToExpect.com	Get Your Baby to Move! • Try these tricks for getting your baby-to-be to kick, flip and roll it	10/4/2016
Pottery Barn Kids	Hurry, hurry! 🎉 Day Premier Event + FREE shipping end tonight! • In stores & online: Ho	10/4/2016
Pottery Barn Kids	100s of deals. 1 day only. ALL our brands. GO! (Plus, free shipping!) • Today only, in store	10/4/2016
WhatToExpect.com	Enter to win top baby feeding essentials • Win up to \$1,500 worth of products! Four luck	10/4/2016
WhatToExpect.com	9 Facts About Your Vagina • Surprising and fascinating facts about a body part you thought	10/3/2016
Pottery Barn Kids	Surprise! Get a FREE treat bag when you shop at your local store • Plus, at your local stor	10/3/2016
Huggies	How to change your baby's life • Watch what a hug can do. Please add Huggies@enews-h	10/3/2016
Pottery Barn Kids	Last day! 🧸 Nursery savings end tonight (cribs, diaper bags and more!) • And today only	10/3/2016
WhatToExpect.com	5 Baby Items I Regret Buying • Save money, time and space by reconsidering items this mc	10/2/2016
Pottery Barn Kids	This just in 🌲 NEW ARRIVALS for the Holidays • Plus, LAST DAY to SAVE on all Santa bags	10/2/2016
Pottery Barn Kids	Hurry to SAVE on Halloween! 20-40% OFF 'til midnight • Save on Halloween costumes, tr	10/2/2016
Pottery Barn Baby	Your NEW nursery awaits! Get free design advice & more → • For projects large and smal	10/1/2016

Fun with Fine Print

What To Expect privacy policy
4000 words = 8 pages long



TODAY - APR 21

Video: Your Pregnancy Week 7
That tail is gone, and baby's getting even cuter.



Healthier Choices for Your Cravings
For every sweet or salty junk food you're craving now, there's a healthier choice that (coupled wi...



TRACK YOUR PROGRESS

Get personalized daily updates about your pregnancy.



The *What to Expect* 'Select Partners' List

Secure <https://www.whattoexpect.com/select-partners/>

WHAT TO EXPECT

GROUPS NEWS PRECONCEPTION PREGNANCY FIRST YEAR TODDLER FAMILY BABY REGISTRY

OUR PARTNERS A - L

- Armwell
Privacy Policy
- buybuy BABY
Privacy Policy
- Cord Blood Registry
Privacy Policy
- Disney Baby
Privacy Policy
- Enfamil
Privacy Policy
- Freshly Picked
Privacy Policy
- Jet
Privacy Policy
- Kimberly-Clark
Huggies
Privacy Policy

Community
Connect... Get Answers... Share
Explore

Advertisement

Secure <https://www.whattoexpect.com/select-partners/>

WHAT TO EXPECT

GROUPS NEWS PRECONCEPTION PREGNANCY FIRST YEAR TODDLER FAMILY BABY REGISTRY

OUR PARTNERS M - Z

- Minted
Privacy Policy
- Motherhood Maternity
Privacy Policy
- Mustela
Privacy Policy
- My Birth Plan
Privacy Policy
- Philips AVENT
Privacy Policy
- Pottery Barn Kids
Privacy Policy
- Sam's Club
Privacy Policy
- Similac
Privacy Policy

How This Can Go Wrong

Not everyone who gets pregnant stays pregnant.

The Internet Thinks I'm Still Pregnant

Modern Love

By AMY PITTMAN SEPT. 2, 2016



Brian Ross



Fun with Fine Print

The Bump privacy policy
4700 words = 10 pages



Baby's First Wiretap

If you use this feature to call a third party, we will record the phone call and any message you leave for the third party, as well as call information such as the number dialed, the date and time of the call and its duration, and your location as determined by your area code or as otherwise permitted.

WTF

~Contacted The Bump~

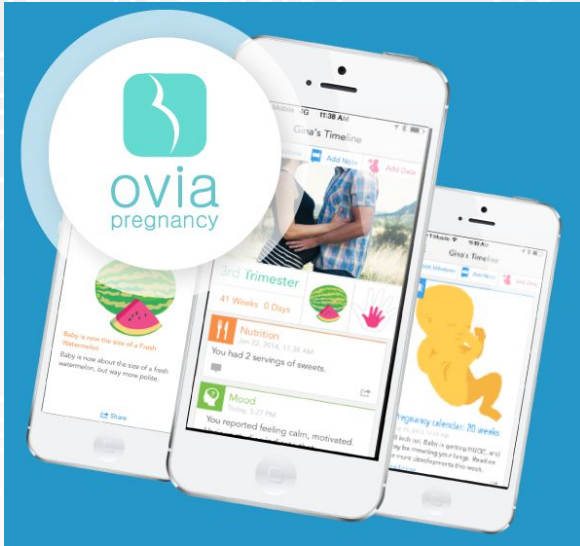


Their response: “[W]e do not record phone calls. The language is legacy language from prior contemplated features for The Knot that we do not use in either The Knot or The Bump apps. I've sent a note to my legal team to update this language in our privacy policy.”

Fun with Fine Print

Ovia Terms of Use

6,100 words = 14 pages



Downloader, beware

“Please do not ask Ovia Health for--or rely on--anything we communicate as medical advice. Although our apps, website, images, content, and communications may reference medical topics, we make no warranty whatsoever that any of the articles are accurate, up to date, or error free. ”

ovia[™]
pregnancy



That sounds a
lot like medical advice



An app as unique as your pregnancy.

Your own daily timeline of articles and personalized feedback on your pregnancy. All your pregnancy questions answered, from "Is this normal?" to "Can I eat this?".

Download for



Real-time alerts when your
symptoms could be dangerous



Track your weight, sleep,
moods, activity, and more



See your baby's actual hand
size



Quick lookups for food,
symptoms and medications

Trust Issues

The Accuracy of Web Sites and Cellular Phone Applications in Predicting the Fertile Window

Setton, Robert MD; Tierney, Christina MD; Tsai, Tony MD

Obstetrics & Gynecology: July 2016 - Volume 128 - Issue 1 - p 58–63

doi: 10.1097/AOG.0000000000001341

Contents: Original Research

RESULTS: “Data from 20 websites and 33 apps were collected. Of all the websites and apps used, one web site and three apps predicted the precise fertile window.”

Their Findings

Table 2. Fertility Windows and Ovulation Dates Predicted by Android Applications Compared With the Average U.S. Fertility Period (Gold Standard)*

Application No.	Predicted Fertile Window (January)	Predicted Date of Ovulation	No. of Days Outside the Fertile Window [†]	No. of Fertile Days Not Included [‡]	% Accuracy [§]
1	13–17	NA	2	3	60
2	10–17	NA	2	0	75
3 [§]	11–14	January 14	0	2	100
4	13–17	January 15	2	3	60
5 [§]	10–16	NA	1	0	86
6 [§]	12–17	January 15	2	2	66.6
7	7–19	January 15	7	0	46
8	10–17	January 15	2	0	75
9	11–17	January 15	2	1	71
10 [§]	10–15	January 15	0	0	100
11 [§]	12–17	NA	2	2	66.6
12 [§]	12–15	January 15	0	2	100
13	11–16	NA	1	1	83
14	8–18	NA	5	0	55
15	10–16	January 15	1	0	86
16	10–16	January 15	1	0	86
17	10–16	January 15	1	0	86
18	10–17	NA	2	0	75
19	11–18	January 15	3	1	63
20 [§]	11–18	January 15	3	1	63

NA, not applicable.

* Gold standard fertility window, January 10–15.

[†] January 10–15.

[‡] Calculated by dividing the total number of correct predicted fertile days by the total number of predicted fertile days, both correct and incorrect.

[§] Duplicate applications in Apple iOS and Android.

The Pregnancy Panopticon



Sorry, Ellev



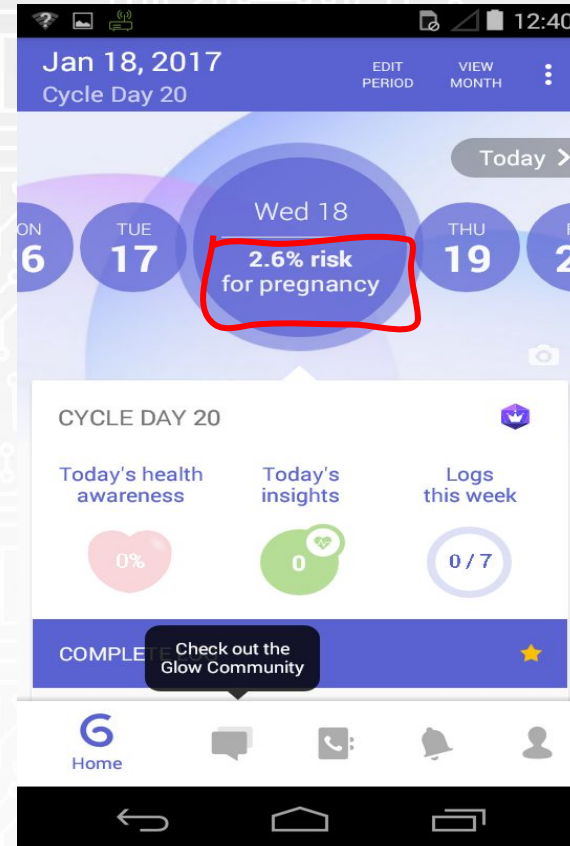
Our Findings

These have got to be the two dumbest people alive



Methods

- Static Analysis
 - JADX APK Decompiler
 - Android Studio
- Dynamic Analysis
 - MITM Proxy
- Kryptowire



Methods - JADX & Studio



Methods - MITM Proxy



Methods - Kryptowire

Kryptowire EMM Portal							
							
APPLICATION ANALYSIS							
Analyzed Apps							
Submit Android App							
SUPPORT TICKET							
LOGOUT							
App	Platform	Date Submitted	Threat Score	Security Issues	Reports		
 Pregnancy Calendar Ver: 2.0.2		2017-06-28 16:41:33	81.4	Can write to external storage, </> Loads code dynamically, ! Requests excessive permissions, \$ Has in app purchases, Integrates with social network, Makes a connection to foreign country	PDF Report	NIAP HTML	JSON
 Pregnancy + Ver: 4.2.2		2017-06-28 14:55:03	81.1	Can access camera, Can access microphone, Can access contacts, Can send with SMS messages, Can write to external storage, </> Loads code dynamically, ! Requests excessive permissions, \$ Has in app purchases, Integrates with social network, Makes a connection to foreign country	PDF Report	NIAP HTML	JSON
 Pink Pad Pro Ver: 3.9.8		2017-06-28 14:56:15	78.6	Can write to external storage, </> Loads code dynamically, Can obtain user/device specific information, ! Requests excessive permissions, \$ Has in app purchases, Integrates with social network, Exposes sensitive information, Makes a connection to foreign country, Integrates with ad network	PDF Report	NIAP HTML	JSON
 WebMD Baby Ver: 2.3.4		2017-05-16 12:35:19	78.6	Can access camera, Can access contacts, Can write to external storage, </> Loads code dynamically, ! Requests excessive permissions, Integrates with social network, Exposes sensitive information, Makes a connection to foreign country, Integrates with ad network	PDF Report	NIAP HTML	JSON
 Ovia Pregnancy Ver: 2.0.0		2017-06-28 15:06:44	77.2	Can write to external storage, ! Requests excessive permissions, Integrates with social network, Exposes sensitive information, Makes a connection to foreign country	PDF Report	NIAP HTML	JSON
 Get Baby Ver: 1.1.2		2017-06-28 14:59:04	75.1	Can write to external storage, </> Loads code dynamically, ! Requests excessive permissions, Integrates with social network, Exposes sensitive information, Integrates with ad network	PDF Report	NIAP HTML	JSON
 Eve Ver: 2.7.3		2017-04-26 08:51:40	74.6	Can access contacts, Can write to external storage, </> Loads code dynamically, ! Requests excessive permissions, \$ Has in app purchases, Integrates with social network, Makes a connection to foreign country	PDF Report	NIAP HTML	JSON
 BabyBump Pro Ver: 6.2.8		2017-06-28 15:05:46	72.7	Can write to external storage, </> Loads code dynamically, Can obtain user/device specific information, ! Requests excessive permissions, Integrates with social network, Makes a connection to foreign country	PDF Report	NIAP HTML	JSON
 First Response Ver: 1.0.0		2017-06-28 15:00:00	60.4	Can access microphone, Can access contacts, Can write to external storage, </> Loads code dynamically, ! Requests excessive permissions, Integrates with social network, Makes a connection to foreign country	PDF Report	NIAP HTML	JSON

Methods - Kryptowire



Android Application Analysis

BabyBump Pregnancy Pro

Version: 6.2.8



Report Created: 06-28-2017 16:03 PM
SHA1: 77b5bcb8e0603b005fdb8ef6bba2fef91717b9e
MD5: 1dda1df27b0aaa6410908405a9afad90

[Expand All](#)

APPLICATION INFORMATION

Package Name: com.alt12.babybump	Min SDK: 11
Version Name: 6.2.8	Target SDK: 25
Version Code: 6028	SHA1: 77b5bcb8e0603b005fdb8ef6bba2fef91717b9e
File Size: 18.47MB	MD5: 1dda1df27b0aaa6410908405a9afad90

PLAY STORE INFORMATION

Category: Health & Fitness	Content Rating: Mature 17+
Downloads: 1,000,000 - 5,000,000	Last Updated: June 20, 2017
User Rating: 4.55	Number of Ratings: 25430
Developer: Alt12 Apps, Inc.	Developer Email: support-babybumpandroid@alt12.com

HIGH RISK ANALYSIS FINDINGS

1



- 1 User password exposed

MEDIUM RISK ANALYSIS FINDINGS

5



- Creates resources accessible from outside parties
- Can write to the external storage
- Requested excessive permissions
- Dynamically loads a Java class
- Makes a possible connection to foreign country

LOW RISK ANALYSIS FINDINGS

6



- Possibly integrates with a social network
- Can access the Internet
- Can access general location
- Access to list of accounts from Account Manager
- Can read the external storage
- Gets the unique ID of the device

Code Execution and Content Injection

- HTML sent over plaintext and rendered directly.
- MITM Attacker could easily execute arbitrary javascript.

Account Hijacking



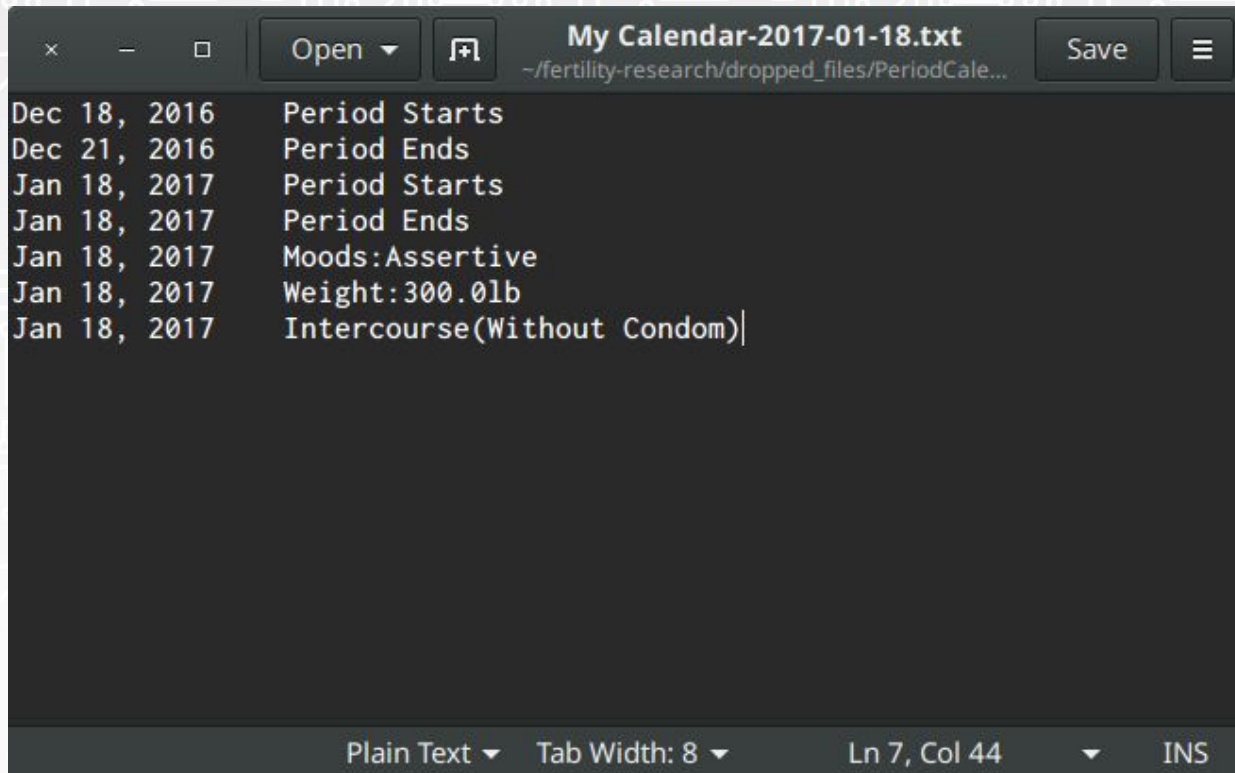
- Pinkpad
- WebMD Baby
- My Calendar
- The Bump

Personal Information Leaks



- Why does pinkpad send my location to the server every time it starts?
- Email, Name, Gender, Pregnancy status, etc.

Look at this fucking text file



The screenshot shows a text editor window titled "My Calendar-2017-01-18.txt" with a file path of "~/fertility-research/dropped_files/PeriodCale...". The window contains a list of dates and corresponding events. The status bar at the bottom indicates "Plain Text", "Tab Width: 8", "Ln 7, Col 44", and "INS" mode.

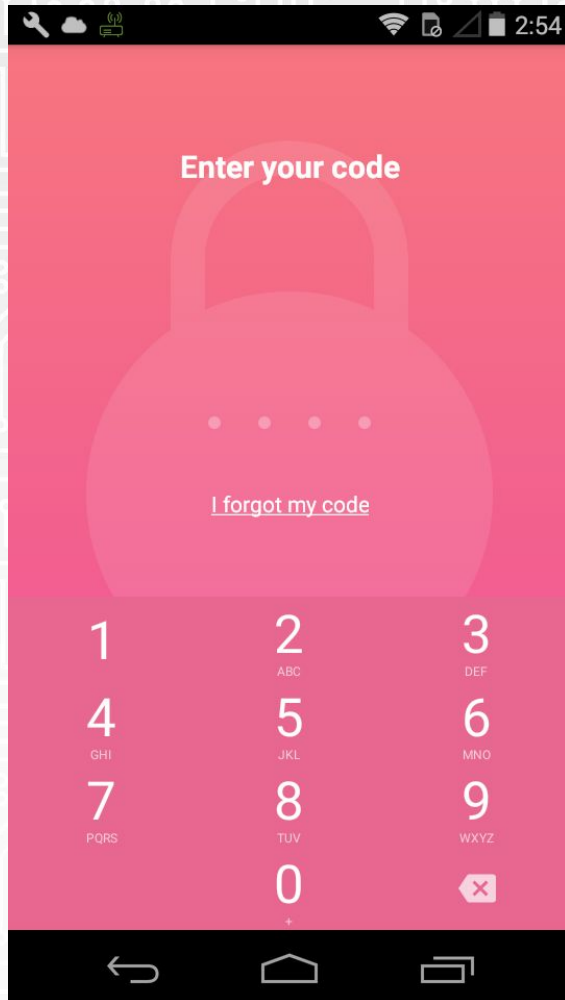
```
Dec 18, 2016    Period Starts
Dec 21, 2016    Period Ends
Jan 18, 2017    Period Starts
Jan 18, 2017    Period Ends
Jan 18, 2017    Moods:Assertive
Jan 18, 2017    Weight:300.0lb
Jan 18, 2017    Intercourse(Without Condom)|
```

Third Party Tracking



Pin Locks

- 4 character limit
- Trivial to brute force
- No protection against root access
- “I forgot my code...”
- Probably shouldn't rely on this



Files Not Deleted

- The Bump
- Upload pictures of pregnancy progress, ultrasounds, baby photos, etc.
- Once you delete the pictures...
- They aren't really deleted but simply unlinked from your account. Still available on the public internet.

Permissions

- Location - Glow, WTE, Preg+, WebMD, Pinkpad, Baby Bump, Ovia Pregnancy, Ovia Fertility, Maya
- Contacts - Eve, Preg+, WebMD, BabyBump, Ovia Pregnancy
- Device ID - Glow, Eve, WTE, Pinkpad, BabyBump, The Bump
- Phone - Glow, Preg+, TheBump, Baby Bump
- SMS - Preg+

Certificate Pinning!



- Glow, Nurture, Eve, Clue all implement certificate pinning.
- My bank doesn't even do this!
- Seems a little extra though...

Vendor Response



Vendor Response

- We contacted all of the vendors that had security problems
- WTE, WebMD, Preg+, Alt12, MyCalendar (both), Glow, Ptracker
- We received a response back from pTracker and Glow, who fixed the issues.
- Everyone else ignored or sent a form letter.

More Vendor Response

Kash heard back from everyone she emailed about privacy and policy issues, except for Everyday Health Inc. (What To Expect app) and alt12. Companies pay attention to journalists... or at least better understand what we're telling them about.

What Can Hackers Do

- There are real threat models that none of these apps protect against.
- We need to convince app writers to take security and privacy issues more seriously.

These apps were useful for my
first pregnancy...
But the privacy tradeoffs might not
be worthwhile

Thanks!

- Thanks to Kryptowire for donating their analysis services.
- Thanks to Dave Choffnes and Jingjing Ren for help with Recon analysis.
- Thanks to Gizmodo Media and EFF for supporting this research.
- Thanks to Defcon, Nikita, and the Goons!
- Thanks to Ellev for inspiring this research!

Questions?

Kashmir Hill

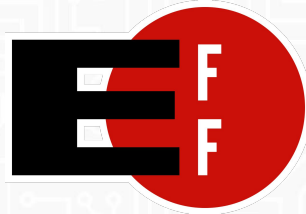
kashmir.hill@gizmodomedia.com - @kashhill

Cooper Quintin

cooperq@eff.org - @cooperq

<https://jezebel.com/what-happens-when-you-tell-the-internet-youre-pregnant-1794398989>

<https://www.eff.org/wp/pregnancy-panopticon>



GIZMODO